- **Factor the greatest common factor from a polynomial.**
  Step 1. Find the GCF of all the terms of the polynomial.
  Step 2. Rewrite each term as a product using the GCF.
  Step 3. Use the Distributive Property 'in reverse' to factor the expression.
  Step 4. Check by multiplying the factors.

Exercises

Review Exercises

Add and Subtract Polynomials

Identify Polynomials, Monomials, Binomials and Trinomials

In the following exercises, determine if each of the following polynomials is a monomial, binomial, trinomial, or other polynomial.

494. $y^2 + 8y - 20$

495. $-6a^4$

496. $9x^3 - 1$

497. $n^3 - 3n^2 + 3n - 1$

Determine the Degree of Polynomials

In the following exercises, determine the degree of each polynomial.

498. $16x^2 - 40x - 25$

499. $5m + 9$

500. -15

501. $y^2 + 6y^3 + 9y^4$

Add and Subtract Monomials

In the following exercises, add or subtract the monomials.

502. $4p + 11p$

503. $-8y^3 - 5y^3$

504. Add $4n^5$, $-n^5$, $-6n^5$

505. Subtract $10x^2$ from $3x^2$

Add and Subtract Polynomials

In the following exercises, add or subtract the polynomials.

506. $(4a^2 + 9a - 11) + (6a^2 - 5a + 10)$

507. $(8m^2 + 12m - 5) - (2m^2 - 7m - 1)$

508. $(y^2 - 3y + 12) + (5y^2 - 9)$

509. $(5u^2 + 8u) - (4u - 7)$

510. Find the sum of $8q^3 - 27$ and $q^2 + 6q - 2$

511. Find the difference of $x^2 + 6x + 8$ and $x^2 - 8x + 15$

Evaluate a Polynomial for a Given Value of the Variable

In the following exercises, evaluate each polynomial for the given value.

512. $200x - \frac{1}{5}x^2$ when $x = 5$

513. $200x - \frac{1}{5}x^2$ when $x = 0$

514. $200x - \frac{1}{5}x^2$ when $x = 15$

515. $5 + 40x - \frac{1}{2}x^2$ when $x = 10$

516. $5 + 40x - \frac{1}{2}x^2$ when $x = -4$

517. $5 + 40x - \frac{1}{2}x^2$ when $x = 0$

518. A pair of glasses is dropped off a bridge 640 feet above a river. The polynomial $-16t^2 + 640$ gives the height of the glasses t seconds after they were dropped. Find the height of the glasses when $t = 6$.

519. The fuel efficiency (in miles per gallon) of a bus going at a speed of x miles per hour is given by the polynomial $-\frac{1}{160}x^2 + \frac{1}{2}x$. Find the fuel efficiency when $x = 20$ mph.

Use Multiplication Properties of Exponents

Simplify Expressions with Exponents

In the following exercises, simplify.

520. 6^3

521. $\left(\frac{1}{2}\right)^4$

522. $(-0.5)^2$

523. -3^2

Simplify Expressions Using the Product Property of Exponents

In the following exercises, simplify each expression.

524. $p^3 \cdot p^{10}$

525. $2 \cdot 2^6$

526. $a \cdot a^2 \cdot a^3$

527. $x \cdot x^8$

Simplify Expressions Using the Power Property of Exponents

In the following exercises, simplify each expression.

528. $(y^4)^3$

529. $(r^3)^2$

530. $(3^2)^5$

531. $(a^{10})^y$

Simplify Expressions Using the Product to a Power Property

In the following exercises, simplify each expression.

532. $(8n)^2$

533. $(-5x)^3$

534. $(2ab)^8$

535. $(-10mnp)^4$

Simplify Expressions by Applying Several Properties

In the following exercises, simplify each expression.

536. $(3a^5)^3$

537. $(4y)^2(8y)$

538. $(x^3)^5(x^2)^3$

539. $(5st^2)^3(2s^3t^4)^2$

Multiply Monomials

In the following exercises, multiply the monomials.

540. $(-6p^4)(9p)$

541. $\left(\frac{1}{3}c^2\right)\left(30c^8\right)$

542. $(8x^2y^5)(7xy^6)$

543. $\left(\frac{2}{3}m^3n^6\right)\left(\frac{1}{6}m^4n^4\right)$

Multiply Polynomials

Multiply a Polynomial by a Monomial

In the following exercises, multiply.

544. $7(10-x)$

545. $a^2(a^2-9a-36)$

546. $-5y(125y^3-1)$

547. $(4n-5)(2n^3)$

Multiply a Binomial by a Binomial

In the following exercises, multiply the binomials using various methods.

548. $(a+5)(a+2)$

549. $(y-4)(y+12)$

550. $(3x+1)(2x-7)$

551. $(6p-11)(3p-10)$

552. $(n+8)(n+1)$

553. $(k+6)(k-9)$

554. $(5u-3)(u+8)$

555. $(2y-9)(5y-7)$

556. $(p+4)(p+7)$

557. $(x-8)(x+9)$

558. $(3c+1)(9c-4)$

559. $(10a-1)(3a-3)$

Multiply a Trinomial by a Binomial

In the following exercises, multiply using any method.

560. $(x+1)(x^2-3x-21)$

561. $(5b-2)(3b^2+b-9)$

562. $(m+6)(m^2-7m-30)$

563. $(4y-1)(6y^2-12y+5)$

Divide Monomials

Simplify Expressions Using the Quotient Property of Exponents

In the following exercises, simplify.

564. $\frac{2^8}{2^2}$

565. $\frac{a^6}{a}$

566. $\frac{n^3}{n^{12}}$

567. $\frac{x}{x^5}$

Simplify Expressions with Zero Exponents

In the following exercises, simplify.

568. 3^0

569. y^0

570. $(14t)^0$

571. $12a^0-15b^0$

Simplify Expressions Using the Quotient to a Power Property

In the following exercises, simplify.

572. $\left(\frac{3}{5}\right)^2$

573. $\left(\frac{x}{2}\right)^5$

574. $\left(\frac{5m}{n}\right)^3$

575. $\left(\frac{s}{10t}\right)^2$

Simplify Expressions by Applying Several Properties

In the following exercises, simplify.

576. $\frac{(a^3)^2}{a^4}$

577. $\frac{u^3}{u^2 \cdot u^4}$

578. $\left(\frac{x}{x^9}\right)^5$

579. $\left(\frac{p^4 \cdot p^5}{p^3}\right)^2$

580. $\frac{(n^5)^3}{(n^2)^8}$

581. $\left(\frac{5s^2}{4t}\right)^3$

Divide Monomials

In the following exercises, divide the monomials.

582. $72p^{12} \div 8p^3$

583. $-26a^8 \div \left(2a^2\right)$

584. $\frac{45y^6}{-15y^{10}}$

585. $\frac{-30x^8}{-36x^9}$

586. $\frac{28a^9b}{7a^4b^3}$

587. $\frac{11u^6v^3}{55u^2v^8}$

588. $\frac{\left(5m^9n^3\right)\left(8m^3n^2\right)}{\left(10mn^4\right)\left(m^2n^5\right)}$

589. $\frac{42r^2s^4}{6rs^3} - \frac{54rs^2}{9s}$

Integer Exponents and Scientific Notation

Use the Definition of a Negative Exponent

In the following exercises, simplify.

590. 6^{-2}

591. $(-10)^{-3}$

592. $5 \cdot 2^{-4}$

593. $(8n)^{-1}$

Simplify Expressions with Integer Exponents

In the following exercises, simplify.

594. $x^{-3} \cdot x^9$

595. $r^{-5} \cdot r^{-4}$

596. $\left(uv^{-3}\right)\left(u^{-4}v^{-2}\right)$

597. $\left(m^5\right)^{-1}$

598. $\left(k^{-2}\right)^{-3}$

599. $\frac{q^4}{q^{20}}$

600. $\frac{b^8}{b^{-2}}$

601. $\frac{n^{-3}}{n^{-5}}$

Convert from Decimal Notation to Scientific Notation

In the following exercises, write each number in scientific notation.

602. $5{,}300{,}000$

603. 0.00814

604. The thickness of a piece of paper is about 0.097 millimeter.

605. According to www.cleanair.com, U.S. businesses use about $21{,}000{,}000$ tons of paper per year.

Convert Scientific Notation to Decimal Form

In the following exercises, convert each number to decimal form.

606. 2.9×10^4

607. 1.5×10^8

608. 3.75×10^{-1}

609. 9.413×10^{-5}

Multiply and Divide Using Scientific Notation

In the following exercises, multiply and write your answer in decimal form.

610. $\left(3 \times 10^7\right)\left(2 \times 10^{-4}\right)$

611. $\left(1.5 \times 10^{-3}\right)\left(4.8 \times 10^{-1}\right)$

612. $\frac{6 \times 10^9}{2 \times 10^{-1}}$

613. $\frac{9 \times 10^{-3}}{1 \times 10^{-6}}$

Introduction to Factoring Polynomials

Find the Greatest Common Factor of Two or More Expressions

In the following exercises, find the greatest common factor.

614. $5n,\ 45$

615. $8a,\ 72$

616. $12x^2,\ 20x^3,\ 36x^4$

617. $9y^4,\ 21y^5,\ 15y^6$

Factor the Greatest Common Factor from a Polynomial

In the following exercises, factor the greatest common factor from each polynomial.

618. $16u - 24$

619. $15r + 35$

620. $6p^2 + 6p$

621. $10c^2 - 10c$

622. $-9a^5 - 9a^3$

623. $-7x^8 - 28x^3$

624. $5y^2 - 55y + 45$

625. $2q^5 - 16q^3 + 30q^2$

Practice Test

626. For the polynomial $8y^4 - 3y^2 + 1$
ⓐ Is it a monomial, binomial, or trinomial?
ⓑ What is its degree?

In the following exercises, simplify each expression.

627. $(5a^2 + 2a - 12) + (9a^2 + 8a - 4)$

628. $(10x^2 - 3x + 5) - (4x^2 - 6)$

629. $\left(-\frac{3}{4}\right)^3$

630. $n \cdot n^4$

631. $(10p^3q^5)^2$

632. $(8xy^3)(-6x^4y^6)$

633. $4u(u^2 - 9u + 1)$

634. $(s + 8)(s + 9)$

635. $(m + 3)(7m - 2)$

636. $(11a - 6)(5a - 1)$

637. $(n - 8)(n^2 - 4n + 11)$

638. $(4a + 9b)(6a - 5b)$

639. $\frac{5^6}{5^8}$

640. $\left(\frac{x^3 \cdot x^9}{x^5}\right)^2$

641. $(47a^{18}b^{23}c^5)^0$

642. $\frac{24r^3s}{6r^2s^7}$

643. $\frac{8y^2 - 16y + 20}{4y}$

644. $(15xy^3 - 35x^2y) \div 5xy$

645. 4^{-1}

646. $(2y)^{-3}$

647. $p^{-3} \cdot p^{-8}$

648. $\frac{x^4}{x^{-5}}$

In the following exercises, factor the greatest common factor from each polynomial.

649. $80a^3 + 120a^2 + 40a$

650. $-6x^2 - 30x$

651. According to www.cleanair.org, the amount of trash generated in the US in one year averages out to 112,000 pounds of trash per person. Write this number in scientific notation.

652. Convert 5.25×10^{-4} to decimal form.

In the following exercises, simplify, and write your answer in decimal form.

653. $\left(2.4 \times 10^{8}\right)\left(2 \times 10^{-5}\right)$

654. $\frac{9 \times 10^{4}}{3 \times 10^{-1}}$

655. A hiker drops a pebble from a bridge 240 feet above a canyon. The polynomial $-16t^2 + 240$ gives the height of the pebble t seconds a after it was dropped. Find the height when $t = 3$.